

# LIMSLite — Pharma LIMS

Seed Data Setup & Manual Validation Guide

Version 1.0 | 2026-06-03 | For Engineer Reference

SEPARATE GUIDE • SEED DATA + MANUAL VALIDATION

|                |                                                                                               |
|----------------|-----------------------------------------------------------------------------------------------|
| Document ID    | LIMS-SEED-VAL-001                                                                             |
| Purpose        | Seed database setup guide + manual validation steps for each LIMS module                      |
| Companion Doc  | LIMSLite-Testing-Guide.pdf (base 22-step guide)                                               |
| Repo           | <a href="https://github.com/hari4872/Pharma-LIMS">https://github.com/hari4872/Pharma-LIMS</a> |
| Prepared by    | WebSynergies (S) Pte Ltd — Engineering Team                                                   |
| Effective Date | 2026-06-03                                                                                    |

**How to use this guide:** Follow Part A (Seed Data Setup) first — this loads all prerequisite test data into the PostgreSQL database automatically via EF Core migrations. Then work through Part B (Manual Validation) to verify each module end-to-end. Each section lists the exact seeded records to use so there is no ambiguity about test data.



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## A1 System Prerequisites & Stack

Before seeding — verify these are in place

Verify all prerequisites before running the seed. The application will not start or seed correctly if any of these are missing.

| Component   | Required Version | Verify Command                   | Notes                                     |
|-------------|------------------|----------------------------------|-------------------------------------------|
| .NET SDK    | 8.x              | dotnet --version                 | Must be 8.x; 9.x not supported            |
| Node.js     | 18+              | node --version                   | For frontend dev server                   |
| npm         | 9+               | npm --version                    | Bundled with Node.js                      |
| PostgreSQL  | 16 (Neon)        | Connection string in appsettings | Cloud-hosted Neon DB                      |
| EF Core CLI | 8.x              | dotnet ef --version              | Install: dotnet tool install -g dotnet-ef |

### Connection String Location

```
appsettings.json – Connection String
File: backend/src/LIMS.API/appsettings.json
"ConnectionStrings": {
  "DefaultConnection": "Host=;Database=limsdb;
  Username=;Password=;
  SslMode=Require"
}
```

### Start the Application

```
start.bat equivalent commands
# Option 1 – Double-click launcher (Windows)
D:\Pharma-LIMS\start.bat
# Option 2 – Manual
cd D:\Pharma-LIMS\backend
dotnet run --project src/LIMS.API --urls http://localhost:5204
# Frontend (separate terminal)
cd D:\Pharma-LIMS\frontend
npm run dev
# URLs
Frontend : http://localhost:5173
Backend  : http://localhost:5204
Swagger  : http://localhost:5204/swagger
```

## A2 Database Seed Overview

What gets seeded automatically via EF Core migrations

Seed data is applied automatically when **dotnet ef database update** runs. All seed inserts use **ON CONFLICT DO NOTHING** — they are fully idempotent and safe to re-run. The seeds are split across two EF Core migration files:

| Migration File                 | Applied    | What it seeds                                                                                                                                                                  |
|--------------------------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SeedData_CompletePharmWorkflow | 2026-05-13 | Lab, Users, E-sigs, Materials, Instruments, Methods, Spec Templates, Lots, Samples (6), Test Executions (4), Logbook entries (6)                                               |
| SeedData_FullEndToEnd          | 2026-05-22 | Training records (9), Checkpoints (4), Checkpoint params (6), Trigger logs (12), Results reviews (4), COA lines (5), Stability pulls (4), Delivery orders (2), Validation logs |

### Seed Data Flow

- Migration runs top-to-bottom; each seed block depends on the previous
- All foreign keys are satisfied within the same migration file
- Serial sequences are reset after each block to avoid ID collisions
- Re-running migrations is safe — no duplicate data will be inserted

### Run Seed Command

```
EF Core – Apply Migrations + Seed
cd D:\Pharma-LIMS\backend

# Apply all pending migrations (includes seed data)
dotnet ef database update --project src/LIMS.Infrastructure \
--startup-project src/LIMS.API

# Confirm migrations applied:
# Expected: all 29 migrations listed as 'Done'
```

A3 Seeded Users & Credentials

5 users across 4 roles — all ready to log in

All seeded users share the **same password** for ease of testing. In production, passwords must be changed immediately.

| Username   | Full Name            | Role       | Email                     | Password | Lab         |
|------------|----------------------|------------|---------------------------|----------|-------------|
| admin      | System Administrator | Admin      | admin@apexpharma.sg       | password | Apex Pharma |
| analyst1   | Dr. Priya Nair       | Analyst    | priya.nair@apexpharma.sg  | password | Apex Pharma |
| analyst2   | Mr. Rajan Mehta      | Analyst    | rajan.mehta@apexpharma.sg | password | Apex Pharma |
| qa_officer | Ms. Siti Rahman      | QA         | siti.rahman@apexpharma.sg | password | Apex Pharma |
| manager    | Dr. Chen Wei         | LabManager | chen.wei@apexpharma.sg    | password | Apex Pharma |

Role Capabilities Summary

| Role       | Key Permissions                                                           |
|------------|---------------------------------------------------------------------------|
| Admin      | Full access — user management, all modules, audit trail export            |
| Analyst    | Register samples, record test results, manage logbook entries             |
| QA         | Review/sign-off test executions, manage OOS, release batches, approve COA |
| LabManager | All Analyst + QA actions; approve methods; view compliance panel          |

Training Records (Seeded)

Each analyst is pre-trained on all test methods. ValidUntil = 2028. This allows immediate test execution without training block errors.

| User                       | Method              | Training Date | Valid Until |
|----------------------------|---------------------|---------------|-------------|
| analyst1 (Dr. Priya Nair)  | TM-001 HPLC Assay   | 2026-01-03    | 2028-01-03  |
| analyst1                   | TM-002 Dissolution  | 2026-01-03    | 2028-01-03  |
| analyst1                   | TM-003 Karl Fischer | 2026-01-03    | 2028-01-03  |
| analyst2 (Mr. Rajan Mehta) | TM-001 HPLC Assay   | 2026-01-04    | 2028-01-04  |
| analyst2                   | TM-002 Dissolution  | 2026-01-04    | 2028-01-04  |
| analyst2                   | TM-003 Karl Fischer | 2026-01-04    | 2028-01-04  |
| qa_officer                 | TM-001 HPLC Assay   | 2026-01-05    | 2028-01-05  |
| manager                    | TM-001 HPLC Assay   | 2026-01-05    | 2028-01-05  |
| manager                    | TM-003 Karl Fischer | 2026-01-05    | 2028-01-05  |

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Seeded Master Data

Lab, materials, instruments, methods, spec templates

Laboratory

| LabId | Lab Name                 | Location                        | Type |
|-------|--------------------------|---------------------------------|------|
| 1     | Apex Pharma Laboratories | Singapore Science Park, Block 3 | QC   |

Materials

| MaterialId | Name                       | Type            | UOM | Shelf Life |
|------------|----------------------------|-----------------|-----|------------|
| 1          | Paracetamol API            | FinishedProduct | kg  | 730 days   |
| 2          | Ibuprofen API              | FinishedProduct | kg  | 730 days   |
| 3          | Microcrystalline Cellulose | RawMaterial     | kg  | 365 days   |
| 4          | Purified Water             | Reagent         | L   | 1 day      |

Instruments

| InstrumentId | Code     | Name                       | Type        | Status    | Next Cal. Due |
|--------------|----------|----------------------------|-------------|-----------|---------------|
| 1            | HPLC-001 | Agilent 1260 HPLC          | HPLC        | Available | 2026-07-01    |
| 2            | DISS-001 | Distek 2100C Dissolution   | Dissolution | Available | 2026-08-01    |
| 3            | KF-001   | Mettler Toledo KF Titrator | KarlFischer | Available | 2026-09-01    |
| 4            | BAL-001  | Mettler Toledo XS205       | Balance     | Available | 2026-06-15    |
| 5            | HPLC-002 | Waters Alliance e2695      | HPLC        | Available | 2026-07-15    |

Test Methods

| MethodId | Code   | Name                    | Version | Status   |
|----------|--------|-------------------------|---------|----------|
| 1        | TM-001 | HPLC Assay Method       | v2.1    | Approved |
| 2        | TM-002 | Dissolution Test Method | v1.3    | Approved |
| 3        | TM-003 | Karl Fischer Moisture   | v1.0    | Approved |

Specification Templates

| SpecId | Product                  | Test Method         | Parameters (Limits)                                |
|--------|--------------------------|---------------------|----------------------------------------------------|
| 1      | Paracetamol 500mg Tablet | TM-001 HPLC Assay   | Assay% (98–102%), RSD% (<2.0%), Impurity A (<0.1%) |
| 2      | Paracetamol 500mg Tablet | TM-002 Dissolution  | Dissolution 45min (>80%), 60min (>90%)             |
| 3      | Ibuprofen 400mg Tablet   | TM-001 HPLC Assay   | Assay% (98–102%), RSD% (<2.0%)                     |
| 4      | Paracetamol 500mg Tablet | TM-003 Karl Fischer | Moisture% (<0.5%)                                  |

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Seeded Workflow Records

Samples, executions, checkpoints, logbook entries

Material Lots (6 seeded)

| LotId | Lot Number   | Material                   | Qty    | Status     | Expiry     |
|-------|--------------|----------------------------|--------|------------|------------|
| 1     | LOT-2026-001 | Paracetamol API            | 500 kg | Available  | 2028-01-01 |
| 2     | LOT-2026-002 | Ibuprofen API              | 300 kg | Available  | 2028-06-01 |
| 3     | LOT-2026-003 | Microcrystalline Cellulose | 200 kg | Available  | 2027-01-01 |
| 4     | LOT-2026-004 | Purified Water             | 100 L  | Available  | 2026-01-16 |
| 5     | LOT-2026-005 | Paracetamol API            | 250 kg | Quarantine | 2028-03-01 |
| 6     | LOT-2026-006 | Ibuprofen API              | 150 kg | Expired    | 2025-12-31 |

Samples (6 seeded — one per lifecycle status)

| Sampl<br>eld | Sample No.   | Material                  | Status              | Analyst  | Notes                                           |
|--------------|--------------|---------------------------|---------------------|----------|-------------------------------------------------|
| 1            | SMP-2026-001 | Paracetamol API (LOT-001) | Released            | analyst1 | Full lifecycle complete; COA generated          |
| 2            | SMP-2026-002 | Ibuprofen API (LOT-002)   | InTesting           | analyst2 | Currently being tested — use for test execution |
| 3            | SMP-2026-003 | Paracetamol API (LOT-001) | PendingQARevie<br>w | analyst1 | Ready for QA sign-off                           |
| 4            | SMP-2026-004 | Ibuprofen API (LOT-002)   | Registered          | —        | Newly registered; no testing started            |
| 5            | SMP-2026-005 | Paracetamol API (LOT-005) | Rejected            | analyst2 | Failed assay — OOS raised                       |
| 6            | SMP-2026-006 | Ibuprofen API (LOT-006)   | Expired             | —        | Lot expired before testing                      |

Test Executions (4 seeded)

| ExecId | Sample       | Method             | Status         | Instrument | Analyst  |
|--------|--------------|--------------------|----------------|------------|----------|
| 1      | SMP-2026-001 | TM-001 HPLC        | Completed      | HPLC-001   | analyst1 |
| 2      | SMP-2026-001 | TM-002 Dissolution | Completed      | DISS-001   | analyst1 |
| 3      | SMP-2026-002 | TM-001 HPLC        | InProgress     | HPLC-002   | analyst2 |
| 4      | SMP-2026-003 | TM-001 HPLC        | PendingSignOff | HPLC-001   | analyst1 |

Checkpoints (4 seeded — all trigger modes)

| Checkpoi<br>ntId | Code   | Trigger Mode | Type    | Time Slots          | Form Template    |
|------------------|--------|--------------|---------|---------------------|------------------|
| 1                | CP-001 | TimeBased    | Single  | 08:00, 14:00, 20:00 | FT-001 (HPLC)    |
| 2                | CP-002 | OperatorScan | Single  | —                   | FT-001 (HPLC)    |
| 3                | CP-003 | ProcessLog   | Grouped | —                   | FT-002 (pH/Cond) |

|   |        |               |        |   |               |
|---|--------|---------------|--------|---|---------------|
| 4 | CP-004 | DispatchEvent | Single | — | FT-001 (HPLC) |
|---|--------|---------------|--------|---|---------------|

Digital Logbook Entries (6 seeded)

| EntryId | Instrument | Parameter         | Value   | Status     | Analyst  |
|---------|------------|-------------------|---------|------------|----------|
| 1       | HPLC-001   | Assay%            | 99.8%   | Signed     | analyst1 |
| 2       | HPLC-001   | RSD%              | 0.45%   | Signed     | analyst1 |
| 3       | DISS-001   | Dissolution 45min | 85.2%   | Signed     | analyst1 |
| 4       | HPLC-002   | Assay%            | 101.2%  | InReview   | analyst2 |
| 5       | BAL-001    | Weight (mg)       | 499.8mg | Signed     | analyst2 |
| 6       | HPLC-001   | Impurity A%       | 0.08%   | Superseded | analyst1 |



## A6 Running the Seed

Step-by-step walkthrough for engineer

Follow these steps in order. Each step must succeed before proceeding.

| Step | Action / Navigation                                                                                                         | Expected Result                                                      |
|------|-----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| 1    | Open terminal; navigate to backend root.                                                                                    | Prompt shows: D:\Pharma-LIMS\backend>                                |
| 2    | Verify EF Core tools installed: dotnet ef --version                                                                         | Output: Entity Framework Core .NET Command-line Tools 8.x.x          |
| 3    | Verify DB connection by running: dotnet ef migrations list --project src/LIMS.Infrastructure --startup-project src/LIMS.API | All 29 migration names listed without connection errors.             |
| 4    | Run migrations + seed: dotnet ef database update --project src/LIMS.Infrastructure --startup-project src/LIMS.API           | Each migration shown with status. Ends with: 'Done.'                 |
| 5    | If any migration fails, check __EFMigrationsHistory table to see which ran. Re-run the same command to resume.              | Idempotent — safe to re-run. Already-applied migrations are skipped. |
| 6    | Start the backend: start.bat or dotnet run                                                                                  | Console shows: <i>Now listening on http://localhost:5204</i>         |
| 7    | Open Swagger: <a href="http://localhost:5204/swagger">http://localhost:5204/swagger</a> — verify API is live.               | Swagger UI loads with all controller endpoints listed.               |
| 8    | Log in via Swagger or UI: POST /api/auth/login with {username:'admin', password:'password'}                                 | JWT token returned. Copy for Swagger Authorize.                      |

■ If the migration fails with 'relation already exists' errors, the database already has tables. Run the seed-only SQL scripts in `tmp-migration-sprint1/` directly against the DB instead.

## A7 Verifying the Seed

SQL queries to confirm data is in place

Run these queries against the Neon PostgreSQL database (via pgAdmin, psql, or any SQL client) to confirm the seed data loaded correctly.

### Migrations applied

```
SELECT COUNT(*) FROM "__EFMigrationsHistory";
```

**Expected:** 29

### Users seeded

```
SELECT username, "Role" FROM users ORDER BY "UserId";
```

**Expected:** 5 rows: admin, analyst1, analyst2, qa\_officer, manager

### Lab seeded

```
SELECT "LabName" FROM laboratories;
```

**Expected:** 'Apex Pharma Laboratories'

### Materials

```
SELECT COUNT(*) FROM materials;
```

**Expected:** 4

### Instruments

```
SELECT COUNT(*) FROM instruments;
```

**Expected:** 5

### Test Methods

```
SELECT COUNT(*) FROM test_methods;
```

**Expected:** 3

### Samples

```
SELECT "SampleNo","Status" FROM samples ORDER BY "SampleId";
```

**Expected:** 6 rows across all statuses

### Checkpoints

```
SELECT "CheckpointCode","TriggerMode" FROM checkpoints;
```

**Expected:** 4 rows: CP-001 to CP-004

### Training Records

```
SELECT COUNT(*) FROM user_training_records;
```

**Expected:** 9

### Logbook Entries

```
SELECT COUNT(*) FROM digital_logbook_entries;
```

**Expected:** 6

B1

## Login & Authentication

Validate all 5 users + account lockout

| Step | Action / Navigation                                                                               | Expected Result                                                                                                                   |
|------|---------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| 1    | Open http://localhost:5173; enter username= <b>admin</b> password= <b>password</b> ; click Login. | JWT issued; redirected to Dashboard. Role badge = 'Admin'.                                                                        |
| 2    | Log out; log in as <b>analyst1</b> / password.                                                    | Dashboard loads with Analyst view. No Admin menu visible.                                                                         |
| 3    | Log out; log in as <b>qa_officer</b> / password.                                                  | QA view loads. Batch Release and OOS Review menus visible.                                                                        |
| 4    | Log out; log in as <b>manager</b> / password.                                                     | Lab Manager view. Compliance Panel accessible.                                                                                    |
| 5    | Log out; enter correct username but <b>wrong password 5 times</b> in a row.                       | After 5th attempt: HTTP 423 response; message: 'Account locked for 30 minutes'. LoginAuditLogs shows 5 Failed + 1 LockedOut rows. |
| 6    | Attempt login with <b>non-existent username</b> .                                                 | HTTP 401 with message 'Invalid credentials'. No account lockout triggered.                                                        |
| 7    | Log in as admin; navigate to Admin → Audit Trail; filter by Action=Login.                         | All login attempts visible with timestamp, IP, username, outcome.                                                                 |

| Check Item                | What to Verify                                                  | Pass / Fail |
|---------------------------|-----------------------------------------------------------------|-------------|
| <b>JWT returned</b>       | Token present in response; expiry = 8 hours                     | [ ]         |
| <b>Role in token</b>      | Decoded JWT contains both ClaimTypes.Role and 'role' claims     | [ ]         |
| <b>Lockout at 5 fails</b> | Account locked after exactly 5 failed attempts                  | [ ]         |
| <b>Audit logged</b>       | Every login attempt (success or fail) in LoginAuditLogs table   | [ ]         |
| <b>Role-based UI</b>      | Menu items differ per role; no admin options visible to Analyst | [ ]         |

B2

## Sample Registration

Register, e-sign SRF, verify status transitions

**Use seeded data:** Log in as **analyst1**. Use **LOT-2026-001** (Paracetamol API, Available) and **SMP-2026-004** (pre-registered sample) for reference.

| Step | Action / Navigation                                                                                            | Expected Result                                                                                                                              |
|------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Navigate to <b>Samples</b> → <b>Register New Sample</b> .                                                      | Registration form opens: Material, Lot No., Quantity, Storage Condition, Priority.                                                           |
| 2    | Select Material = Paracetamol API; Lot = LOT-2026-001; Qty = 10g; submit.                                      | 5 auto-checks run: lot availability, expiry, quantity, material type, lab assignment. All pass. Sample saved with status <b>Registered</b> . |
| 3    | Click <b>E-Sign SRF</b> ; enter password = <b>password</b> ; meaning = 'Sample received and verified'; submit. | Electronic signature inserted. Sample status → <b>PendingTesting</b> . Barcode log entry created.                                            |
| 4    | Attempt registration with <b>LOT-2026-006</b> (Expired lot).                                                   | HTTP 422 returned: ' <i>Lot is expired — registration blocked</i> '. No sample created.                                                      |
| 5    | Attempt e-sign with <b>wrong password</b> .                                                                    | HTTP 422: ' <i>ESIGN_AUTH_FAILED</i> '. Signature not inserted. Sample status unchanged.                                                     |
| 6    | As <b>admin</b> , view Audit Trail for the newly registered sample.                                            | Audit shows: CREATE event, e-sign event, status change — all with timestamps and user.                                                       |

| Check Item                     | What to Verify                                                            | Pass / Fail |
|--------------------------------|---------------------------------------------------------------------------|-------------|
| <b>5-point auto-check</b>      | All 5 checks (lot, expiry, qty, type, lab) evaluated on POST /samples     | [ ]         |
| <b>Barcode log</b>             | Entry in barcode_logs table with SampleId, timestamp, action='Registered' | [ ]         |
| <b>E-sig immutability</b>      | Signed SRF cannot be edited; edit attempt returns 422 INVALID_STATE       | [ ]         |
| <b>Status = PendingTesting</b> | After e-sign, sample.Status = 'PendingTesting' in DB                      | [ ]         |
| <b>Expired lot block</b>       | POST /samples with expired lot returns 422, no DB row created             | [ ]         |

## B3

## Test Execution &amp; Work Queue

Assign, start, record results, trigger OOS

**Use seeded data:** Sample **SMP-2026-002** is *InTesting* with Execution ID 3 (analyst2, HPLC-002, InProgress). Sample **SMP-2026-004** (Registered) can be used to test full flow from assignment.

| Step | Action / Navigation                                                                            | Expected Result                                                                                    |
|------|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| 1    | Log in as <b>manager</b> ; navigate to <b>Work Queue</b> .                                     | Work Queue shows all pending test executions grouped by priority and due date.                     |
| 2    | Assign SMP-2026-004 (Registered) to analyst1 with Method TM-001, Instrument HPLC-001.          | Assignment saved; analyst1 notified; execution status = <b>Assigned</b> .                          |
| 3    | Log in as <b>analyst1</b> ; open the assigned execution; click <b>Start Testing</b> .          | Status → <b>InProgress</b> ; started_at timestamp set server-side.                                 |
| 4    | Enter parameter results: Assay% = 99.5, RSD% = 0.8, Impurity A = 0.06.                         | Results saved; all within spec; pass_fail = 'PASS' per parameter.                                  |
| 5    | Enter an OOS result: Assay% = <b>95.0</b> (below 98% min limit).                               | is_oos=true on result row; OOS investigation auto-created; execution status → <b>OOS Pending</b> . |
| 6    | Attempt to assign an instrument with status <b>Out of Calibration</b> .                        | HTTP 422: ' <i>INSTRUMENT_OOC</i> '. Assignment blocked.                                           |
| 7    | Attempt to assign analyst with expired training (set valid_until to past date in DB for test). | HTTP 422: ' <i>TRAINING_EXPIRED</i> '. Assignment blocked.                                         |
| 8    | Complete valid execution; click <b>Submit for QA Review</b> .                                  | Status → <b>PendingSignOff</b> ; QA notified.                                                      |

| Check Item                    | What to Verify                                                            | Pass / Fail |
|-------------------------------|---------------------------------------------------------------------------|-------------|
| <b>Server-side started_at</b> | started_at set by backend, not client — cannot be manipulated             | [ ]         |
| <b>OOS auto-creation</b>      | When is_oos=true, complaints_deviations row auto-inserted with type='OOS' | [ ]         |
| <b>Instrument OOC block</b>   | HPLC-002 with CalibrationDue=past → 422 on assignment                     | [ ]         |
| <b>Training expiry block</b>  | analyst with valid_until < today → 422 TRAINING_EXPIRED                   | [ ]         |
| <b>Result calculation</b>     | calculated_result from ParameterCalculationService matches manual calc    | [ ]         |

## B4

## Digital Logbook

Entries, amendments, CSV export — 21 CFR §11 validation

**Use seeded data:** 6 logbook entries pre-loaded (IDs 1–6). Entry 6 has status *Superseded* — use to verify amendment chain.

| Step | Action / Navigation                                                                      | Expected Result                                                                                                                  |
|------|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| 1    | Navigate to <b>Digital Logbook</b> ; verify 6 seeded entries visible.                    | Entries listed with instrument, parameter, value, status badges.                                                                 |
| 2    | Click <b>+ New Entry</b> ; select HPLC-001; enter parameter=Assay%; value=100.1; submit. | Entry created with status <b>Pending</b> ; analyst_id recorded; created_at set by DB DEFAULT NOW().                              |
| 3    | E-sign the entry (password='password'; meaning='Verified by analyst').                   | Entry status → <b>Signed</b> ; electronic_signatures row inserted with full_name, signed_at, meaning.                            |
| 4    | Attempt to edit a <b>Signed</b> entry directly.                                          | HTTP 422: 'INVALID_STATE — cannot edit Signed entry'.                                                                            |
| 5    | Click <b>Amend</b> on the signed entry; enter amendment reason; submit new value.        | Original entry status → <b>Superseded</b> ; superseded_by_id set to new entry ID. New entry created as <b>Pending</b> .          |
| 6    | Attempt to amend a <b>Pending</b> entry (not yet signed).                                | HTTP 422: 'INVALID_STATE — can only amend Signed entries'.                                                                       |
| 7    | Export logbook to CSV; open file.                                                        | CSV contains: EntryId, Instrument, Parameter, Value, Status, AnalystId, FullName, SignedAt, Meaning — all §11.50 fields present. |
| 8    | Check that entry 6 (Superseded) shows full amendment chain in UI.                        | UI shows original value crossed out + pointer to new entry; both visible in audit.                                               |

| Check Item                     | What to Verify                                                        | Pass / Fail |
|--------------------------------|-----------------------------------------------------------------------|-------------|
| <b>DB-level timestamp</b>      | created_at set by DEFAULT NOW() at DB level — client cannot supply it | [ ]         |
| <b>Signed entry immutable</b>  | No UPDATE/DELETE allowed on Signed rows — only via amendment          | [ ]         |
| <b>Amendment chain intact</b>  | superseded_by_id forms a linked chain; no orphan superseded entries   | [ ]         |
| <b>CSV §11.50 fields</b>       | All 9 required fields present in CSV export                           | [ ]         |
| <b>Analyst ID on every row</b> | analyst_id FK present and non-null on every logbook entry             | [ ]         |

B5

## Checkpoints

Validate all 4 trigger modes

**Use seeded data:** 4 checkpoints (CP-001 to CP-004) and 12 trigger log entries already in DB.

| Step | Action / Navigation                                                                        | Expected Result                                                                              |
|------|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| 1    | Navigate to <b>Checkpoints</b> panel; verify CP-001 to CP-004 listed.                      | All 4 checkpoints shown with trigger mode badges and parameter counts.                       |
| 2    | <b>CP-001 TimeBased:</b> At scheduled time (08:00/14:00/20:00), system fires checkpoint.   | Trigger log entry created automatically; analyst receives notification to record results.    |
| 3    | Record results for CP-001: Assay%=99.8, RSD%=0.5. Submit.                                  | Results saved; pass/fail determined against spec limits; status = Pass.                      |
| 4    | <b>CP-002 OperatorScan:</b> Navigate to Scan Entry; scan barcode of a sample.              | Trigger log entry created with TriggerMode='OperatorScan'; checkpoint form opens.            |
| 5    | <b>CP-003 ProcessLog:</b> Complete a process log entry — checkpoint fires automatically.   | Trigger created with TriggerMode='ProcessLog'; grouped parameters (pH + Conductivity) shown. |
| 6    | <b>CP-004 DispatchEvent:</b> Dispatch a delivery order — checkpoint fires on dispatch.     | Trigger created with TriggerMode='DispatchEvent'; DeliveryOrder field populated.             |
| 7    | Enter an out-of-spec result on any checkpoint.                                             | OOS flag raised on the trigger log; QA notified; checkpoint status = Fail.                   |
| 8    | Verify seeded offline-sync entry (TriggerId=8, IsOfflineSync=true) visible in trigger log. | Entry shows IsOfflineSync badge; demonstrates Annex 11 §4.3 offline sync capability.         |

B6

## QA Review & Batch Release

Sign-off, COA generation, batch release

**Use seeded data:** Sample **SMP-2026-003** is *PendingQAReview*. Execution ID 4 is *PendingSignOff*.

| Step | Action / Navigation                                                                      | Expected Result                                                                                                 |
|------|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| 1    | Log in as <b>qa_officer</b> ; navigate to <b>QA Review</b> .                             | Execution ID 4 (SMP-2026-003) shown in queue with status <i>PendingSignOff</i> .                                |
| 2    | Open execution; review all parameter results; click <b>Sign Off</b> .                    | E-sig dialog: enter password='password'; meaning='QA Reviewed and Approved'. Execution status → <b>Signed</b> . |
| 3    | Attempt sign-off as <b>analyst1</b> (non-QA role).                                       | HTTP 403: sign-off endpoint requires QA or LabManager role.                                                     |
| 4    | After sign-off, navigate to <b>Batch Release</b> for SMP-2026-003.                       | Batch Release form shows: sample, all executions, open OOS count (0), release decision.                         |
| 5    | Select Release Decision = <b>Released</b> ; enter release remarks; e-sign.               | Sample status → <b>Released</b> . COA auto-generated. Batch release record created.                             |
| 6    | Select Release Decision = <b>Rejected</b> for SMP-2026-005 (already seeded as Rejected). | Sample status = Rejected; rejection reason stored; COA not generated.                                           |
| 7    | Download COA for the released sample.                                                    | PDF contains: sample no., product, all test results with spec limits, QA signature, release date.               |
| 8    | Attempt to release a sample with an open OOS investigation.                              | HTTP 422: ' <i>Open OOS investigation must be closed before release</i> '.                                      |

| Check Item                 | What to Verify                                                              | Pass / Fail |
|----------------------------|-----------------------------------------------------------------------------|-------------|
| <b>QA-only sign-off</b>    | POST /test-executions/{id}/sign-off returns 403 for Analyst role            | [ ]         |
| <b>E-sig on release</b>    | Release requires valid password verification — not just button click        | [ ]         |
| <b>COA content</b>         | COA PDF includes sample ID, product, results table, spec limits, signatures | [ ]         |
| <b>OOS gate on release</b> | Release blocked if any linked OOS is still Open or Under Investigation      | [ ]         |
| <b>Audit trail</b>         | COA generation event captured in audit trail with QA username + timestamp   | [ ]         |



B7

## OOS Investigation

Phase 1 / Phase 2 workflow and disposition

**Use seeded data:** An OOS was auto-created when SMP-2026-005 was rejected. Use it, or trigger a new OOS by entering Assay% = 95.0 on any test execution.

| Step | Action / Navigation                                                     | Expected Result                                                                             |
|------|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| 1    | Navigate to <b>Quality</b> → <b>OOS Investigations</b> .                | OOS list shows all open and closed investigations with priority and phase badges.           |
| 2    | Open the OOS for SMP-2026-005; verify Phase = Phase 1, Status = Open.   | Phase 1 form: Lab findings, initial root cause hypothesis, analyst notes.                   |
| 3    | Enter Phase 1 findings; upload a supporting document; save.             | Findings saved; document attached; status = Under Investigation.                            |
| 4    | Escalate to Phase 2: click <b>Escalate to Full Investigation</b> .      | Phase 2 section unlocks: Root Cause Category, Root Cause Narrative, CAPA required checkbox. |
| 5    | Select Root Cause = Process; enter narrative; tick CAPA Required; save. | Root cause saved; CAPA record auto-created and linked; QA Manager notified.                 |
| 6    | Select Disposition = <b>Reject Batch</b> ; e-sign; save.                | Sample status confirmed Rejected; OOS status → Closed; final report PDF generated.          |
| 7    | Attempt to close OOS without entering root cause.                       | Validation error: <i>'Root cause is required before closing'</i> .                          |
| 8    | Verify closed OOS is read-only.                                         | All fields greyed out; no edit buttons visible; PDF export still available.                 |

B8

**Quality Events / CAPA**

Event creation, corrective action, effectiveness check

| Step | Action / Navigation                                                                | Expected Result                                                                          |
|------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| 1    | Navigate to <b>Quality</b> → <b>Quality Events</b> ; click <b>+ New Event</b> .    | Form: Type (Deviation/Complaint/Audit Finding), Source, Description, Severity, Priority. |
| 2    | Create a Major Deviation linked to SMP-2026-002; save.                             | Event created (QE-YYYY-NNNNN); status = Open; 72-hour SLA timer starts.                  |
| 3    | Click <b>Create CAPA</b> from the event.                                           | CAPA form: Corrective Action, Preventive Action, Owner, Due Date (default +14 days).     |
| 4    | Assign CAPA to analyst1; due date = today + 14 days; save.                         | CAPA saved and linked to QE; analyst1 notified.                                          |
| 5    | As analyst1, complete the corrective action; upload evidence; submit for review.   | CAPA status → Pending Effectiveness Check.                                               |
| 6    | As qa_officer, schedule effectiveness check; enter result (Effective); close CAPA. | CAPA status → Closed; linked QE also auto-closes.                                        |
| 7    | Attempt to close QE with open CAPA.                                                | Blocked: <i>'All linked CAPAs must be closed first'</i> .                                |
| 8    | Check Compliance Panel for updated CAPA count.                                     | Open CAPA counter on dashboard decrements by 1 after closure.                            |

B9

## Stability Studies

Protocol, time point pulls, OOT detection

**Use seeded data:** 4 stability pull records already seeded (T0 and T3 for Paracetamol and Ibuprofen studies).

| Step | Action / Navigation                                                                                      | Expected Result                                                              |
|------|----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| 1    | Navigate to <b>Stability</b> → <b>Protocols</b> ; verify 2 seeded protocols present.                     | Protocols listed: Paracetamol (25°C/60%RH) and Ibuprofen (40°C/75%RH).       |
| 2    | Create a new stability protocol: product = Paracetamol; condition = 5°C±3°C; time points = T0,T3,T6,T12. | Protocol saved as Draft; time point calendar auto-generated.                 |
| 3    | Submit for QA approval.                                                                                  | Status → Pending Approval; QA Manager notified.                              |
| 4    | As qa_officer, approve the protocol.                                                                     | Status → Active; T0 pull due immediately.                                    |
| 5    | Enter T0 results for all parameters (Assay, Dissolution, pH).                                            | Results saved with T=0; pass/fail against initial spec.                      |
| 6    | Enter T3 results with a downward trend (not OOS but below T0 by >5%).                                    | OOT flag raised (amber); QA notified for trend review; trend chart updated.  |
| 7    | Use ICH Q1A regression endpoint: GET /api/stability/regression/{studyId}                                 | Returns: slope, intercept, predictedShelfLifeMonths, flag (Pass/Warn/Fail).  |
| 8    | Generate stability summary report.                                                                       | PDF: all time points, values, spec limits, trend chart, statistical summary. |

B10

## Compliance Panel & Audit Trail

KPI metrics validation + 21 CFR Part 11

| Step | Action / Navigation                                                                  | Expected Result                                                                                              |
|------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| 1    | Log in as <b>manager</b> ; navigate to <b>Compliance Panel</b> .                     | Dashboard cards: Open OOS, Open CAPA, Training Compliance %, Batch Release Rate.                             |
| 2    | Verify <b>Open OOS Count</b> matches Quality → OOS list (status=Open).               | Counts identical. Discrepancy = bug.                                                                         |
| 3    | Verify <b>Training Compliance %</b> = 100% (all 5 seeded users have valid training). | If < 100%, check user_training_records.valid_until values.                                                   |
| 4    | Navigate to <b>Audit Trail</b> (Admin role required).                                | Full log with Timestamp, User, Module, Action, Before, After columns.                                        |
| 5    | Perform any data change (e.g., edit sample remark); check audit trail immediately.   | New audit entry appears within 5 seconds; Before/After values shown.                                         |
| 6    | Search audit trail: User = analyst1, Date = today.                                   | Only today's analyst1 actions shown.                                                                         |
| 7    | Attempt to delete an audit trail entry via UI or API.                                | No delete button present; DELETE /audit endpoint returns 405 Method Not Allowed.                             |
| 8    | Export audit trail to CSV; verify all \$11.50 fields present.                        | CSV columns: Id, Timestamp (UTC), UserId, Username, Module, Action, RecordId, OldValue, NewValue, IPAddress. |

| Check Item               | What to Verify                                                      | Pass / Fail |
|--------------------------|---------------------------------------------------------------------|-------------|
| INSERT-only audit        | No DELETE or UPDATE on audit_logs table — enforced by DB trigger    | [ ]         |
| UTC timestamps           | All audit timestamps in UTC ISO 8601 format                         | [ ]         |
| Before/After capture     | Field-level changes stored as JSON in old_values/new_values columns | [ ]         |
| IP address logged        | IP address captured on every audit entry from request context       | [ ]         |
| Compliance KPIs accurate | Dashboard figures match manual count from underlying data tables    | [ ]         |

## B11

## User Permissions

Role-based access control — UI and API level

| Step | Action / Navigation                                                                      | Expected Result                                                         |
|------|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| 1    | Log in as <b>analyst1</b> ; attempt to open Admin → Users via direct URL /admin/users.   | 403 Forbidden page shown OR redirect to dashboard.                      |
| 2    | As analyst1, call POST /api/auth/login then attempt PATCH /api/users/1/role via Postman. | HTTP 403 returned. Role unchanged. Audit trail records the attempt.     |
| 3    | As analyst1, attempt GET /api/audit-trail.                                               | HTTP 403. Audit trail is Admin/LabManager only.                         |
| 4    | As <b>qa_officer</b> , open Batch Release and sign off a test execution.                 | Both actions succeed — QA role has these permissions.                   |
| 5    | As <b>manager</b> , verify Compliance Panel accessible.                                  | Panel loads without 403 errors.                                         |
| 6    | As <b>admin</b> , deactivate analyst2 account.                                           | analyst2.IsActive = false; analyst2 login attempt → 'Account disabled'. |
| 7    | Re-activate analyst2; verify login works again.                                          | analyst2 can log in; assigned tasks still visible in work queue.        |

## Role → Endpoint Permission Map

| Endpoint                       | Admin | QA / Mgr | Analyst | Lab Analyst |
|--------------------------------|-------|----------|---------|-------------|
| POST /samples                  | Y     | Y        | Y       | N           |
| POST /samples/{id}/e-sign      | Y     | Y        | Y       | N           |
| POST /test-executions/sign-off | Y     | Y        | N       | N           |
| POST /batch-release            | Y     | Y        | N       | N           |
| GET /audit-trail               | Y     | Y        | N       | N           |
| PUT /users/{id}/role           | Y     | N        | N       | N           |
| GET /compliance                | Y     | Y        | N       | N           |

## B12

## Traceability

Forward trace: lot → sample | Backward trace: sample → lot

**Use seeded data:** LOT-2026-001 was used in SMP-2026-001, SMP-2026-003, SMP-2026-004. SMP-2026-001 is fully released with COA.

| Step | Action / Navigation                                                                             | Expected Result                                                                                   |
|------|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| 1    | Navigate to <b>Traceability</b> → <b>Forward Trace</b> ; enter lot number <b>LOT-2026-001</b> . | All 3 samples that used this lot displayed: SMP-2026-001, 003, 004 with statuses.                 |
| 2    | Drill into SMP-2026-001 from the trace result.                                                  | Full sample detail: lot used, all test executions, results, COA link, release info.               |
| 3    | Navigate to <b>Backward Trace</b> ; enter sample number <b>SMP-2026-001</b> .                   | Source lot LOT-2026-001 shown with supplier, receipt date, CoA, current lot status.               |
| 4    | Call GET /api/traceability/{sampleId} via Swagger for SMP-2026-001.                             | Response < 2 seconds (PQ target). Full JSON chain: lot → sample → executions → results → release. |
| 5    | Export traceability tree to PDF for SMP-2026-001.                                               | PDF shows full material genealogy: lot → registration → testing → QA → release → dispatch.        |
| 6    | Verify traceability for SMP-2026-005 (Rejected).                                                | Chain shows: lot → registration → test execution → OOS → rejection. Complete history.             |

| Check Item                        | What to Verify                                                            | Pass / Fail |
|-----------------------------------|---------------------------------------------------------------------------|-------------|
| <b>Forward trace completeness</b> | All 3 samples using LOT-2026-001 returned in forward trace                | [ ]         |
| <b>Backward trace accuracy</b>    | Backward trace of SMP-2026-001 → exactly LOT-2026-001                     | [ ]         |
| <b>Response time &lt; 2s</b>      | GET /traceability/{sampleId} response under 2 seconds (PQ-OQ target)      | [ ]         |
| <b>Rejected sample trace</b>      | Rejected sample trace chain shows OOS and rejection reason                | [ ]         |
| <b>PDF export completeness</b>    | All chain nodes present in PDF: lot, sample, executions, results, release | [ ]         |